

MACTAN CEBU INTL, Philippines

WMO: 986460

Lat: 10.3224N

Lon: 123.9800E

Elev: 9

StdP: 101.21

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.4	23.9	19.9	14.6	26.5	20.8	15.5	26.7	10.1	27.5	9.2	27.3	2.4	0	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.0	33.2	27.0	32.8	26.9	32.2	26.9	28.6	31.3	28.2	30.9	28.0	30.7	4.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.2	30.4	27.6	23.6	30.2	27.1	22.9	30.0	93.0	31.3	91.3	31.0	89.7	30.9	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.3	6.4	21.2	34.6	1.9	0.8	19.9	35.2	18.8	35.7	17.7	36.2	16.4	36.8		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			19.7	29.2	1.5	0.7	18.7	29.8	17.8	30.2	17.0	30.6	15.9	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	26.8	27.1	27.8	28.8	29.3	28.8	28.2	28.5	28.3	28.1	28.0	27.4
	DBStd	1.16	0.88	0.87	0.89	0.88	0.94	1.10	0.95	0.97	0.95	0.86	0.84	0.91
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6602	521	478	551	563	598	563	564	573	548	562	541	539
	CDD18.3	3561	263	244	293	313	340	313	306	314	298	303	291	281
	CDH23.3	40088	2455	2402	3237	3882	4358	3769	3463	3613	3338	3400	3269	2903
	CDH26.7	13942	534	606	1044	1586	1949	1517	1214	1331	1170	1157	1064	768
Wind	WSAvg	2.9	3.7	3.7	3.4	3.0	2.5	2.3	2.6	2.8	2.6	2.5	2.8	3.2
Precipitation	PrecAvg	1761	128	88	67	74	140	186	205	170	187	215	150	154
	PrecMax	2574	441	256	170	260	391	324	369	280	390	400	472	410
	PrecMin	1189	3	9	5	23	28	87	50	44	69	69	32	23
	PrecStd	340	96	67	53	61	81	64	85	60	80	71	99	100
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.0	31.2	32.1	33.2	34.3	34.1	33.0	33.2	33.1	32.6	32.1	31.6
		MCWB	26.2	26.4	25.9	26.5	27.4	27.5	27.2	27.0	27.2	27.3	26.8	26.7
	2%	DB	30.2	30.7	31.5	32.4	33.2	33.0	32.1	32.5	32.2	31.9	31.3	31.0
		MCWB	25.9	26.0	26.0	26.4	26.9	27.1	27.0	26.9	27.0	27.0	26.8	26.6
	5%	DB	29.8	30.1	31.0	32.0	32.8	32.2	31.6	31.9	31.8	31.2	31.0	30.2
		MCWB	25.8	25.7	25.9	26.4	26.9	27.2	26.9	26.9	26.9	26.9	26.8	26.4
	10%	DB	29.2	29.6	30.4	31.4	32.1	31.6	31.0	31.2	31.0	30.9	30.5	29.9
		MCWB	25.6	25.5	25.8	26.4	26.9	27.1	26.9	26.9	26.9	26.9	26.6	26.3
	0.4%	WB	27.7	27.8	28.1	28.3	29.0	29.1	28.7	28.6	28.7	28.8	28.6	28.2
		MCDB	29.3	29.9	30.2	30.9	32.1	32.2	31.5	31.2	31.3	31.0	30.6	30.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.0	27.1	27.4	27.8	28.4	28.4	28.0	28.0	28.1	28.1	27.9	27.5
		MCDB	29.1	29.4	30.0	30.7	31.5	31.4	30.8	30.9	30.9	30.6	30.2	29.7
	5%	WB	26.5	26.5	26.9	27.4	27.9	27.8	27.5	27.5	27.5	27.5	27.4	27.0
		MCDB	28.7	29.0	29.7	30.5	31.1	30.9	30.4	30.6	30.5	30.2	29.9	29.3
	10%	WB	26.1	26.1	26.5	27.0	27.5	27.4	27.1	27.1	27.1	27.1	26.9	26.6
		MCDB	28.4	28.7	29.3	30.3	30.8	30.5	30.0	30.3	30.2	29.9	29.6	28.9
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.2	5.6	5.7	6.0	5.9	5.6	5.9	5.8	5.6	5.1	4.8
		MCDBR	5.5	5.7	6.2	6.2	6.7	6.6	6.3	6.6	6.6	6.2	5.7	5.4
		MCWBR	2.9	2.9	2.7	2.6	2.7	2.8	3.0	3.2	3.1	3.1	2.8	2.7
	5% WB	MCDBR	5.1	5.4	5.6	5.7	6.1	6.2	5.9	6.1	6.1	5.8	5.4	5.1
		MCWBR	3.1	3.2	3.0	2.9	3.1	3.2	3.2	3.4	3.4	3.3	3.1	2.9
Clear-Sky Solar Irradiance	taub		0.436	0.423	0.423	0.428	0.427	0.421	0.423	0.441	0.437	0.450	0.433	0.433
	taud		2.411	2.438	2.445	2.429	2.449	2.476	2.470	2.405	2.413	2.373	2.429	2.424
	Ebn at Noon		872	896	898	883	867	862	863	860	873	860	869	866
	Edh at Noon		118	118	118	119	115	110	111	121	121	124	114	114
All-Sky Solar Radiation	RadAvg		4.00	4.83	5.63	6.18	5.71	5.16	4.87	5.06	5.00	4.83	4.52	4.04
	RadStd		0.52	0.50	0.54	0.31	0.39	0.31	0.30	0.33	0.43	0.30	0.38	0.36

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.24	+0.21	N/A	+0.43	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page